

NETWORKWALKS

WEEK 3

PASSWORD SECURITY

ASSESSMENT

Offline Password Recovery, Browser-Based Dictionary Attacks,
Evidence Correlation & Defensive Analysis

Document Control	Value
Author	Chandrashekar Bala
Program / Batch	Networkwalks Cybersecurity Internship — B083C
Assessment	Week 3 — Password Cracking / Password Security
Primary modules	W3-PM1 — John the Ripper; W3-PM2 — Networkwalks Tools
Artifacts	My Locked PDF1.pdf / PDF2.pdf / PDF3.pdf
Evidence package	Renamed evidence set E01–E25
Document status	Detailed technical assessment — evidence-based

Authorization & handling

All activities documented in this report were performed against supplied laboratory artifacts within the authorized Networkwalks internship context. Recovered passwords are treated as lab credentials and are excluded from public-facing evidence recommendations.

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1. Executive Summary

This assessment documents a controlled password-security exercise using three password-protected PDF laboratory artifacts. The objective was to demonstrate the mechanics of offline password recovery and then reproduce the workflow using the browser-based Networkwalks tools.

The assessment deliberately distinguishes four different claims: (1) a crackable verifier can be extracted, (2) a candidate password can be recovered by a cracking engine, (3) the recovered candidate actually unlocks the source artifact, and (4) the security consequence can be interpreted without overgeneralizing from a small laboratory sample.

Assessment result

W3-PM1 recovered all three supplied PDF credentials with John the Ripper using RockYou. W3-PM2 recovered PDF2 and PDF3 with the Networkwalks built-in dictionary. For PDF1, the Networkwalks built-in dictionary was exhausted without a match; the tool explicitly recommended a larger wordlist.

Artifact	Local JtR / RockYou	Networkwalks built-in list	Observed evidence
PDF1	Recovered — 1.639 g/s in the captured cracking output; separate captured run context also records candidate throughput	Not recovered — 100/100 candidates, 9 pw/s	E04/E05 + E16
PDF2	Recovered — 1.639 g/s, 104.9 p/s	Recovered — 91/100, 9 pw/s	E11 + E17/E18
PDF3	Recovered — 2.380 g/s, 3504 p/s	Recovered — 35/100, 4 pw/s	E12 + E19/E20

The central technical observation is candidate coverage. PDF1 was not recovered by the bounded browser dictionary but was recovered by the larger local RockYou corpus. This demonstrates why a failed dictionary attack is not equivalent to proof of password strength.

2. Scope, Authorization & Rules of Engagement

The Week 3 task material defines PM2 as a password-cracking lab using the Networkwalks Hash Calculator and Networkwalks Password Cracker. The workflow is limited to locked PDFs supplied for the exercise and is intended for learning under authorized conditions.

In scope	Out of scope
Supplied My Locked PDF1.pdf / PDF2.pdf / PDF3.pdf	Third-party credentials or accounts
Offline extraction of PDF password-verification material	Online authentication attacks
John the Ripper with local wordlists	Credential attacks against live services
Networkwalks Hash Calculator	Unrelated targets
Networkwalks Password Cracker	Production security claims not supported by evidence
qpdf validation of recovered lab credentials	Publication of lab passwords in public repositories

Boundary statement

The observed results describe the supplied laboratory corpus only. They do not establish the security posture of Networkwalks production systems, employee accounts, or enterprise password storage.

3. Assessment Objectives

- Demonstrate extraction of a PDF-specific crackable password verifier.
- Identify the correct cracking representation and execution path.
- Recover credentials using a controlled dictionary attack.
- Independently validate at least the demonstrated recovery chain where evidence is available.
- Compare a locally controlled cracking workflow with a browser-based educational workflow.
- Explain why candidate-set coverage affects dictionary attack outcomes.
- Translate the technical observations into defensive password-security controls.
- Preserve evidence in a form suitable for an assessment report and a sanitized form suitable for public portfolio use.

4. Threat Model and Technical Attack Chain

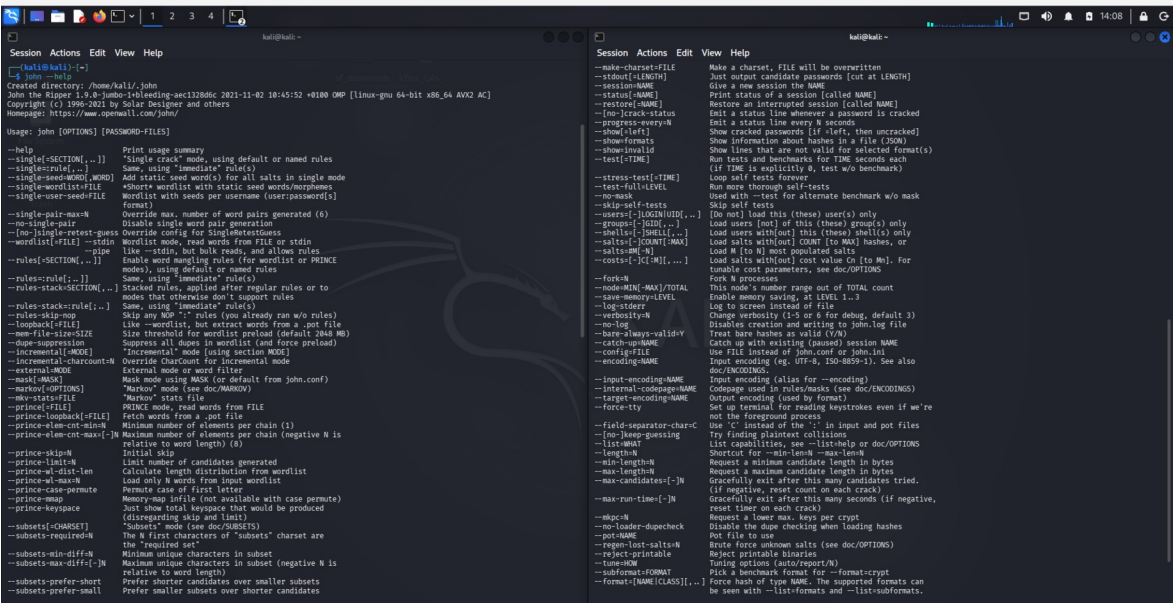
The exercise models an offline attacker who has obtained a protected PDF and can derive a password-verification representation from it. Once that representation exists, the attacker no longer needs to interact with the PDF through an online login interface for every guess.

Stage	Attacker capability	Security control being tested
Artifact acquisition	Obtains protected file	File access control / data exposure
Verifier extraction	Derives \$pdf\$ representation	Protection of credential-verification material
Candidate generation	Chooses dictionary / rules / masks	Password predictability
Offline verification	Tests candidates without lockout	Adaptive hashing / verifier cost
Recovery	Obtains usable password	Password policy + uniqueness + MFA
Post-recovery use	Attempts to unlock artifact	Encryption strength and password reuse controls

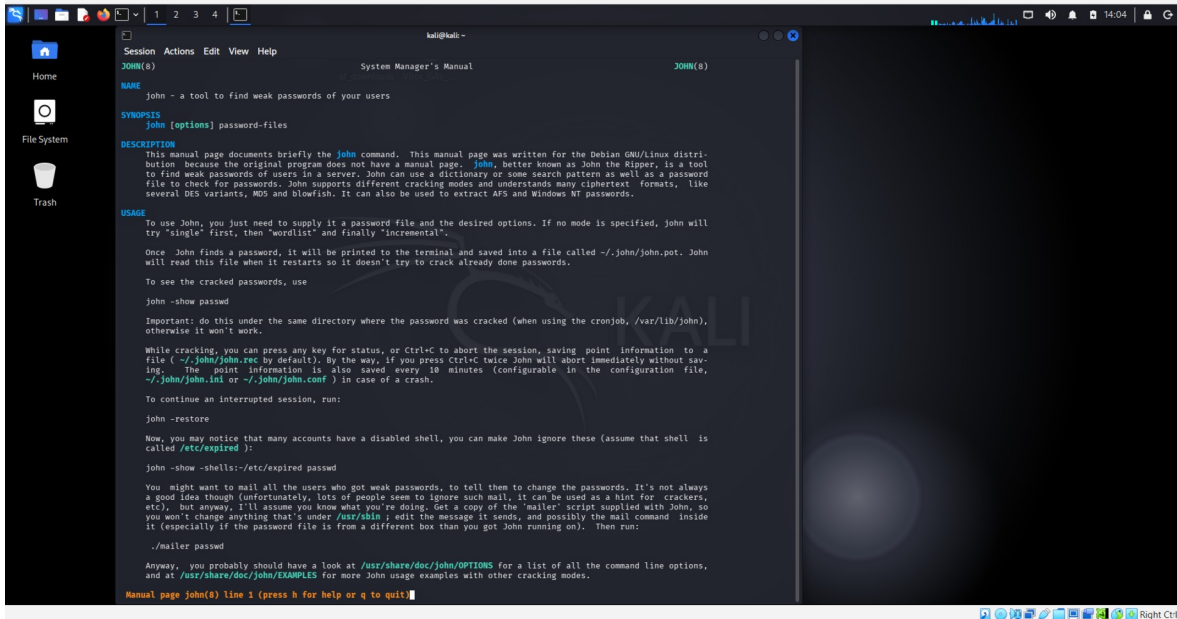
This is why password security cannot be evaluated only through online login controls. Rate limiting can slow an online attacker, but an exposed offline verifier changes the problem into one of password entropy, verifier cost, and attacker compute.

5. Laboratory Environment & Tooling

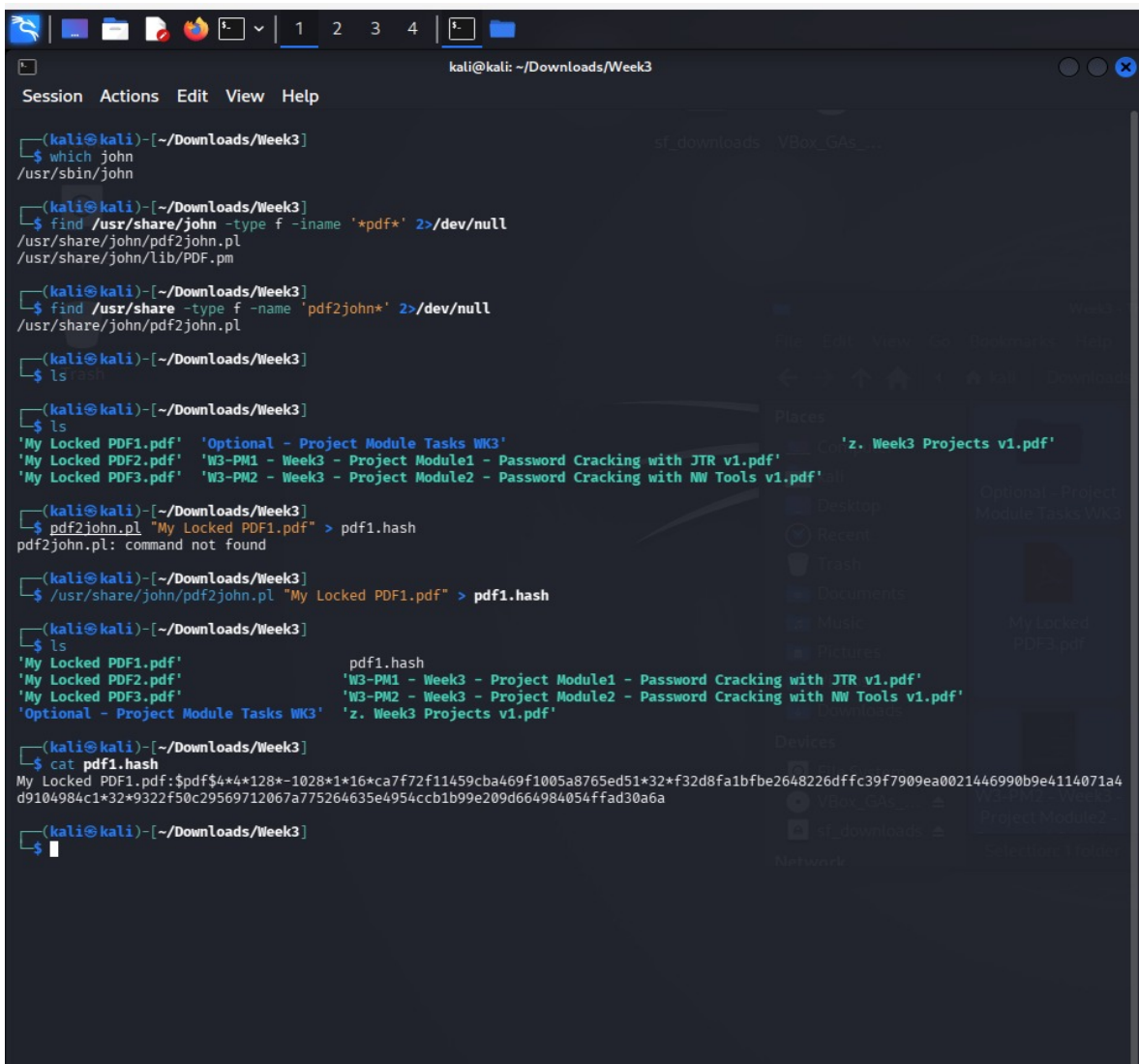
Component	Role in assessment
Kali Linux	Local assessment environment
John the Ripper 1.9.0-jumbo-1	Offline password-recovery engine
pdf2john.pl	PDF password-verifier extraction
RockYou	Large dictionary used for local recovery
qpdf	Independent password acceptance, decryption and PDF validation
pdftinfo / pdftimages / pdftotext / pdftoppm	Decrypted artifact inspection
Networkwalks Hash Calculator	Browser-based PDF hash extraction
Networkwalks Password Cracker	Browser-based dictionary attack



E01 — JtR version and command-line capability context.



E02 — JtR manual evidence describing dictionary/wordlist operation.



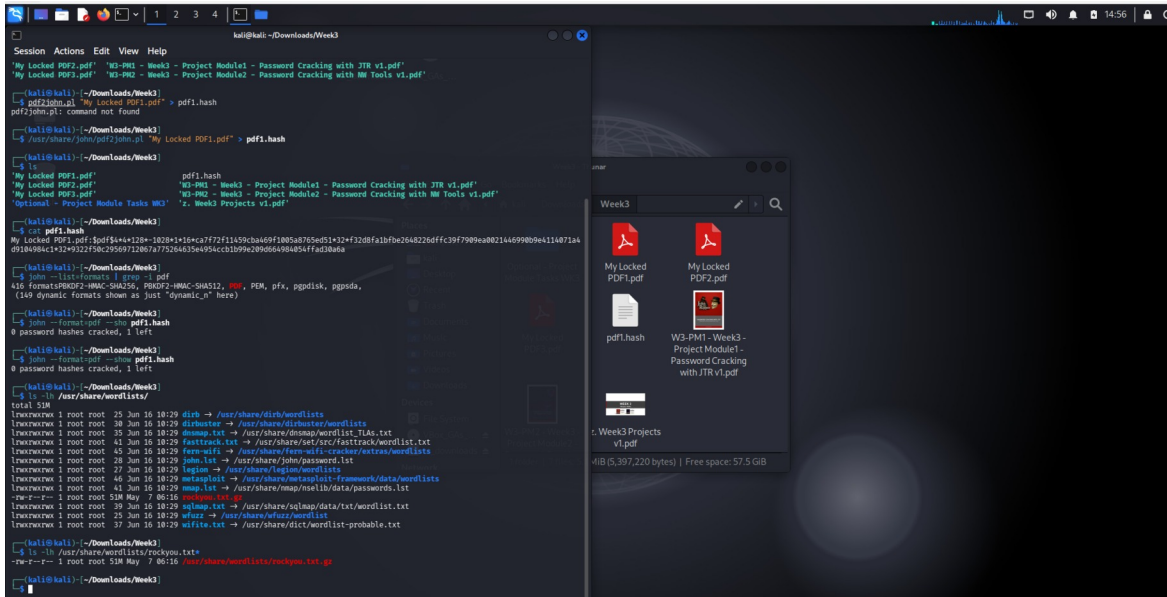
E03 — Discovery of pdf2john.pl and the initial PDF1 extraction workflow.

6. W3-PM1 — John the Ripper

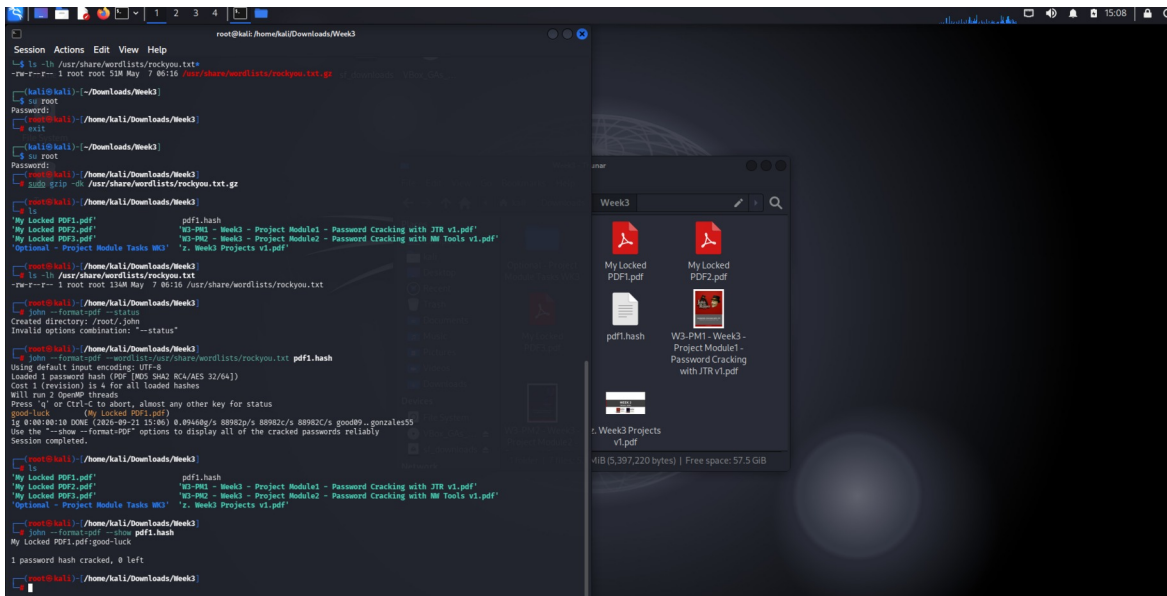
PM1 used a locally controlled offline workflow. The important distinction is that the cracking engine was not asked to interact with a remote service; it was given the extracted PDF verifier and a local candidate corpus.

6.1 PDF1 — Hash extraction and format validation

The evidence shows discovery of the distribution's pdf2john utility, extraction of the PDF1 verifier, and subsequent validation that the resulting representation could be loaded by JtR as a PDF hash.



E04 — PDF1 hash extraction and JtR format validation.



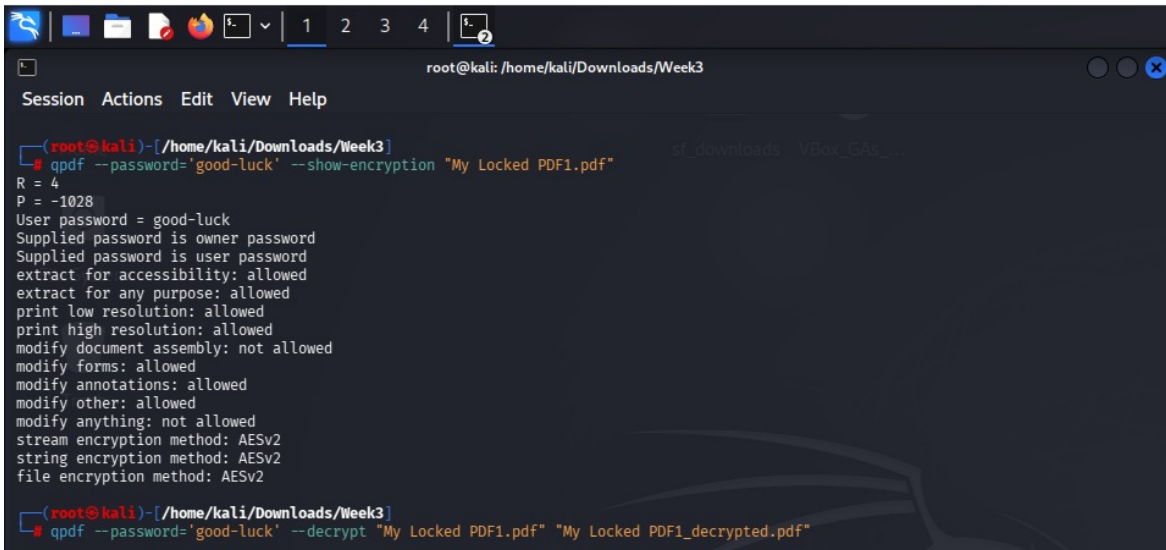
E05 — RockYou preparation and JtR configuration/status context.

6.2 PDF1 — Recovery

The RockYou dictionary attack recovered the laboratory password. JtR reported the PDF format, revision cost 4, and two OpenMP threads. The captured terminal output records an observed cracking rate of approximately 1.639 g/s and 104.9 candidates/s in the displayed recovery run.

6.3 PDF1 — Independent cryptographic validation

Independent validation is the most important evidentiary enhancement over a basic lab submission. qpdf accepted the recovered password and identified AESv2 as the file, stream, and string encryption method.



```

root@kali: /home/kali/Downloads/Week3
Session Actions Edit View Help

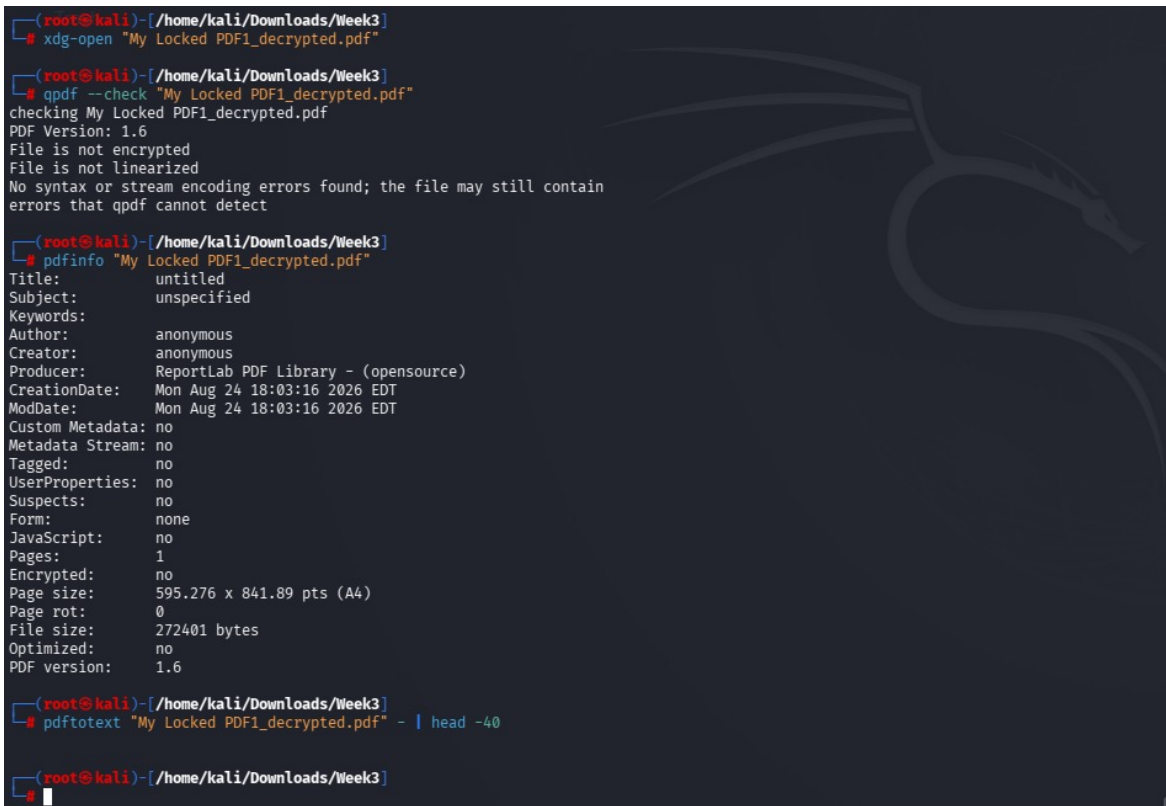
(root@kali)-[/home/kali/Downloads/Week3]
# qpdf --password='good-luck' --show-encryption "My Locked PDF1.pdf"
R = 4
P = -1028
User password = good-luck
Supplied password is owner password
Supplied password is user password
extract for accessibility: allowed
extract for any purpose: allowed
print low resolution: allowed
print high resolution: allowed
modify document assembly: not allowed
modify forms: allowed
modify annotations: allowed
modify other: allowed
modify anything: not allowed
stream encryption method: AESv2
string encryption method: AESv2
file encryption method: AESv2

(root@kali)-[/home/kali/Downloads/Week3]
# qpdf --password='good-luck' --decrypt "My Locked PDF1.pdf" "My Locked PDF1_decrypted.pdf"

```

E08 — qpdf independently accepted the recovered password and reported AESv2.

The PDF was then decrypted and structurally checked. The captured qpdf output reports a valid decrypted PDF with no syntax or stream encoding errors. pdftinfo was used to inspect document metadata; subsequent analysis showed that the document contained an embedded JPEG rather than extractable text.



```

(root@kali)-[/home/kali/Downloads/Week3]
# xdg-open "My Locked PDF1_decrypted.pdf"

(root@kali)-[/home/kali/Downloads/Week3]
# qpdf --check "My Locked PDF1_decrypted.pdf"
checking My Locked PDF1_decrypted.pdf
PDF Version: 1.6
File is not encrypted
File is not linearized
No syntax or stream encoding errors found; the file may still contain
errors that qpdf cannot detect

(root@kali)-[/home/kali/Downloads/Week3]
# pdftinfo "My Locked PDF1_decrypted.pdf"
Title:          untitled
Subject:        unspecified
Keywords:
Author:         anonymous
Creator:        anonymous
Producer:       ReportLab PDF Library - (opensource)
CreationDate:   Mon Aug 24 18:03:16 2026 EDT
ModDate:       Mon Aug 24 18:03:16 2026 EDT
Custom Metadata: no
Metadata Stream: no
Tagged:        no
UserProperties: no
Suspects:      no
Form:          none
JavaScript:    no
Pages:         1
Encrypted:     no
Page size:     595.276 x 841.89 pts (A4)
Page rot:      0
File size:     272401 bytes
Optimized:     no
PDF version:   1.6

(root@kali)-[/home/kali/Downloads/Week3]
# pdftotext "My Locked PDF1_decrypted.pdf" - | head -40

(root@kali)-[/home/kali/Downloads/Week3]
#

```

E09 — PDF1 decryption, qpdf integrity validation, and pdftinfo inspection.

```

root@kali:~/Downloads/Week3
$ pdftotext "My Locked PDF1_decrypted.pdf" - | head -40

root@kali:~/Downloads/Week3
$ pdftimages -list "My Locked PDF1_decrypted.pdf"
page  num  type  width height color comp bpc  enc  interp  object ID x-ppi y-ppi size ratio
1      0  image  1055  1491  rgb    3   8  jpeg   no      7  0   128   128  265K 5.7%

root@kali:~/Downloads/Week3
$ pdftoppm -png -f 1 -singlefile \
  "My Locked PDF1_decrypted.pdf" \
  "My_Locked_PDF1_page1"

root@kali:~/Downloads/Week3
$ ls -lh My_Locked_PDF1_page1.png
-rw-rw-r-- 1 root root 1.5M Sep 21 15:52 My_Locked_PDF1_page1.png

root@kali:~/Downloads/Week3
$ xdg-open My_Locked_PDF1_page1.png

root@kali:~/Downloads/Week3
$

```

E10 — PDF1 content inspection, embedded-image analysis, and rendering.

6.4 PDF2 — Recovery

PDF2 was extracted into JtR format, initially reported as uncracked, and then recovered with the RockYou dictionary. The captured run reports 1.639 g/s and 104.9 p/s/c/s.

```

kali@kali:~/Downloads/Week3
$ cd Week3
$ ls
My Locked PDF1_decrypted.pdf  My Locked PDF2.pdf  pdf1.hash  'z. Week3 Projects v1.pdf'
My Locked PDF1_page1.png    My Locked PDF3.pdf  pdf2.hash  'W3-PM1 - Week3 - Project Module1 - Password Cracking with JTR v1.pdf'
My Locked PDF1.pdf          'Optional - Project Module Tasks WK3'  'W3-PM2 - Week3 - Project Module2 - Password Cracking with HW Tools v1.pdf'

kali@kali:~/Downloads/Week3
$ /usr/share/john/pdft2john.pl "My Locked PDF2.pdf" > pdf2.hash

kali@kali:~/Downloads/Week3
$ ls
My Locked PDF1_decrypted.pdf  My Locked PDF2.pdf  My Locked PDF3.pdf  pdf1.hash  'W3-PM1 - Week3 - Project Module1 - Password Cracking with JTR v1.pdf'  'z. Week3 Projects v1.pdf'
My Locked PDF1_page1.png    My Locked PDF2.pdf  'Optional - Project Module Tasks WK3'  pdf2.hash  'W3-PM2 - Week3 - Project Module2 - Password Cracking with HW Tools v1.pdf'

kali@kali:~/Downloads/Week3
$ cat pdf2.hash
My Locked PDF2.pdf:$pdf$4+*128+*1028*1+10+0053f7cdebf13b1c0f93ed229d3b1ad32*0f113c5aa39ad974364d36a857da76798021446990b9e4114071a4d9184984c1*32*cecdac74b1965a0268803b3524e1374c955cb09cc3c45316494d946ef81af1

kali@kali:~/Downloads/Week3
$ john --format=pdf --show pdf2.hash
0 password hashes cracked, 1 left

kali@kali:~/Downloads/Week3
$ john --format=pdf --wordlist=/usr/share/wordlists/rockyou.txt pdf2.hash
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
1g 0:00:00.00 DONE (2020-09-25 00:46) 1.639g/s 104.9p/s 104.9c/s 123456..charlie
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed

kali@kali:~/Downloads/Week3
$ john --format=pdf --show pdf2.hash
My Locked PDF2.pdf:password1

1 password hash cracked, 0 left

kali@kali:~/Downloads/Week3
$

kali@kali:~/Downloads/Week3
$

kali@kali:~/Downloads/Week3
$

kali@kali:~/Downloads/Week3
$

```

E11 — PDF2 extraction, RockYou recovery, and john --show confirmation.

6.5 PDF3 — Recovery

PDF3 followed the same workflow and was recovered with RockYou. The captured run reports 2.380 g/s and 3504 p/s/c/s.

```

kali@kali:~/Downloads/Week3
$ /usr/share/john/pdft2john.pl "My Locked PDF3.pdf" > pdf3.hash

kali@kali:~/Downloads/Week3
$ ls
My Locked PDF1_decrypted.pdf  My Locked PDF2.pdf  pdf1.hash  'W3-PM1 - Week3 - Project Module1 - Password Cracking with JTR v1.pdf'
My Locked PDF1_page1.png    My Locked PDF3.pdf  pdf2.hash  'W3-PM2 - Week3 - Project Module2 - Password Cracking with HW Tools v1.pdf'
My Locked PDF1.pdf          'Optional - Project Module Tasks WK3'  pdf3.hash  'z. Week3 Projects v1.pdf'

kali@kali:~/Downloads/Week3
$ cat pdf3.hash
My Locked PDF3.pdf:$pdf$4+*128+*1028*1+161634e0542eff4e1b0b32d25ce15a9a7281324b7782bfc9a24fb2f8A506283a8f1b0021446990b9e4114071a4d9184984c1*32*757226e4b552cd57988f5134214b9192809427a6bf550ea94a2995c772ab02

kali@kali:~/Downloads/Week3
$ john --format=pdf --show pdf3.hash
0 password hashes cracked, 1 left

kali@kali:~/Downloads/Week3
$ john --format=pdf --wordlist=/usr/share/wordlists/rockyou.txt pdf3.hash
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
1g 0:00:00.00 DONE (2020-09-25 00:56) 2.380g/s 3504p/s 3504c/s 3504c/s snuggles..iluvu
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed

kali@kali:~/Downloads/Week3
$ john --format=pdf --show pdf3.hash
My Locked PDF3.pdf:1qaz2wsx

1 password hash cracked, 0 left

kali@kali:~/Downloads/Week3
$

```

E12 — PDF3 extraction, RockYou recovery, and john --show confirmation.

7. W3-PM2 — Networkwalks Tools

PM2 reproduced the fundamental workflow through the browser: PDF → Networkwalks Hash Calculator → crackable \$pdf\$ representation → Networkwalks Password Cracker → dictionary candidates.

7.1 PDF1 — Hash Calculator



Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF



TextFilePDF



Click to choose a file, or drop one here

Hashing happens entirely in your browser

My Locked PDF1.pdf (266.9 KB)

Choose a file to hash. Large files are read in your browser and may take a moment.

☐ Uppercase output

MD5

not secure

112330f8d73e9f94b7263404799f4a3b

CopyDownload

SHA-1

not secure

67cb14f88c84c5cd37f45d74631b915aeb8e4468

CopyDownload

SHA-256

CopyDownload

565843cf47c802ea5b7481d5e24c8e99a6a3f8a9eb861ef71bc0b1180be50e4e

SHA-384

CopyDownload

2dfd77f662442ff2e6e0c396a6a749f082cf227c49c52cd1635a08aabbbaa52c389d10589a6864c6ce0dc7f90b14dd8cd

SHA-512

CopyDownload

1f3da52a4449ce8fc25accd414e75b7ca4e110d3b15cf8c5939f2487e238c07ab7d3bf107ea3505cc9c0eda94eb147ef39903762d86a6628c74aef97136e2953

E13 — PDF1 uploaded to the Networkwalks Hash Calculator.



Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF



Text

File

PDF



Click to choose a PDF, or drop one here
Nothing is uploaded, the PDF is parsed locally

My Locked PDF1.pdf | 266.9 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

\$pdf\$4*4*128*-1028*1*16*ca7f72f11459cba469f1005a8765ed51*32*f32d8fa1bfbe2648226dff39f7909ea0021446990b9e4114071a4d9104984c1*32*9322f50c29569712067a775264635e4954ccb1b99e209d664984054ffad30a6a

copy

Download

Revision: R4 Version: V4 Key length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

E14 — Networkwalks extracted a crackable PDF hash and displayed revision v4 / 128-bit key length.

7.2 PDF1 — Password Cracker

networkwalks.com/password-cracker/



Dictionary Attack Lab

Password Cracker through Dictionary Attacks

Hash every word in a wordlist and match it against a PDF password hash, the same idea John the Ripper uses.

1 Extract the \$pdf\$ hash in the [Hash Calculator](#) 2 Paste it below 3 Run the attack, then open your PDF with the password

PDF HASH (\$PDF\$...)

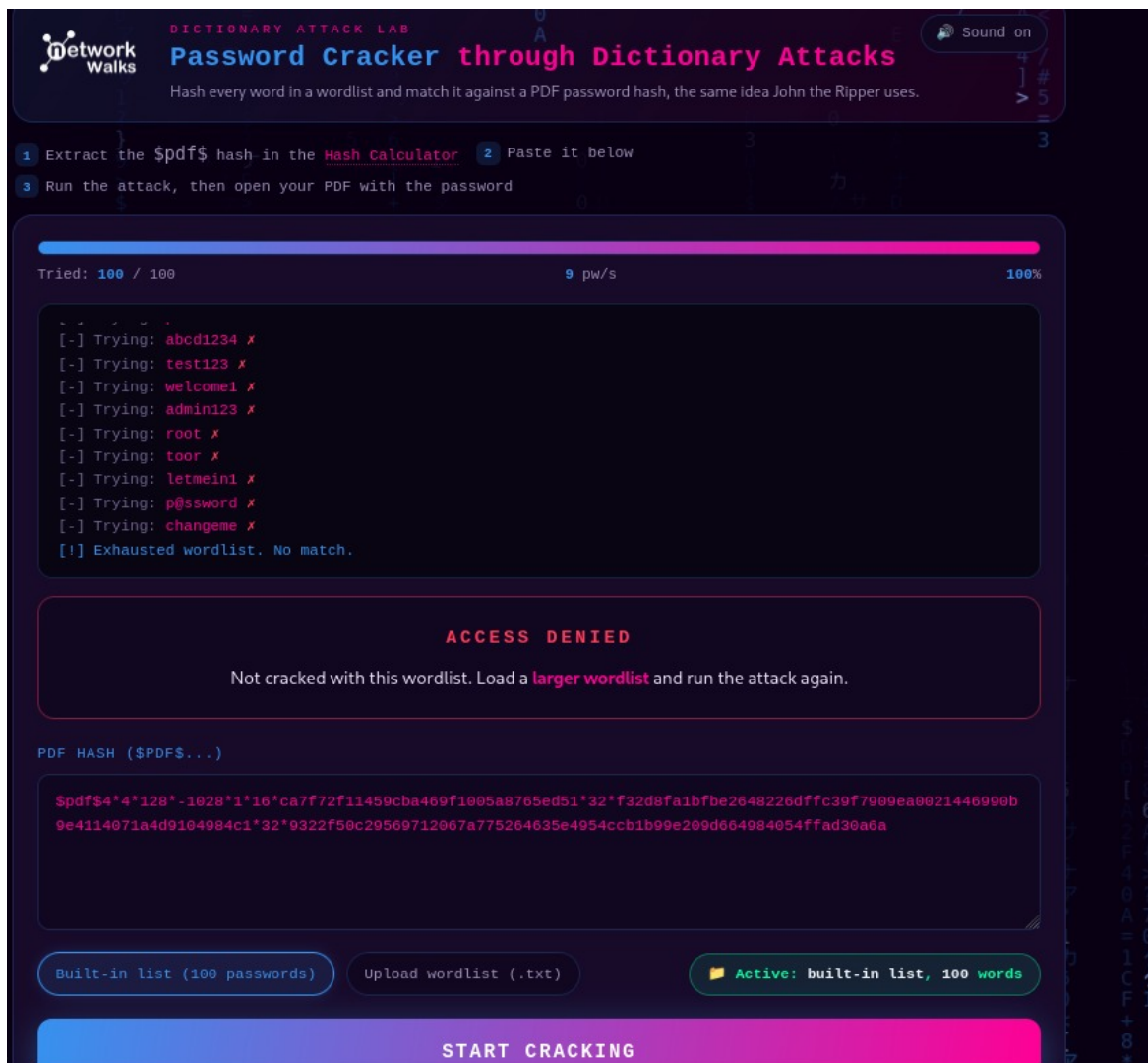
\$pdf\$4*4*128*-1028*1*16*ca7f72f11459cba469f1005a8765ed51*32*f32d8fa1bfbe2648226dff39f7909ea0021446990b9e4114071a4d9104984c1*32*9322f50c29569712067a775264635e4954ccb1b99e209d664984054ffad30a6a

Built-in list (100 passwords) Upload wordlist (.txt) Active: built-in list, 100 words

START CRACKING

Free tool by Networkwalks Academy | Cybersecurity & Ethical Hacking with AI

E15 — PDF1 hash supplied to the Networkwalks Password Cracker.



E16 — Built-in dictionary exhausted without a match; tool recommends a larger wordlist.

Interpretation

This is a bounded negative result, not a statement that PDF1 is secure. The same artifact was recovered locally with RockYou. The evidence therefore isolates candidate coverage as a major variable.

7.3 PDF2 — Hash extraction and recovery



Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF



TextFilePDF



Click to choose a PDF, or drop one here

Nothing is uploaded, the PDF is parsed locally

My Locked PDF2.pdf | 204.7 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

```
$pdf$4*4*128*-1028*1*16*0853f2cde0ef15b1c0f93ed229d3b1ad*32*8f13ce5aa39ad974364d36a057da76790021446990b9e4114071a4d9104984c1*32*ceecdac74b19b5a62688d3b3524e1374c955cbb9cc3c45316494d9446ef81af1
```

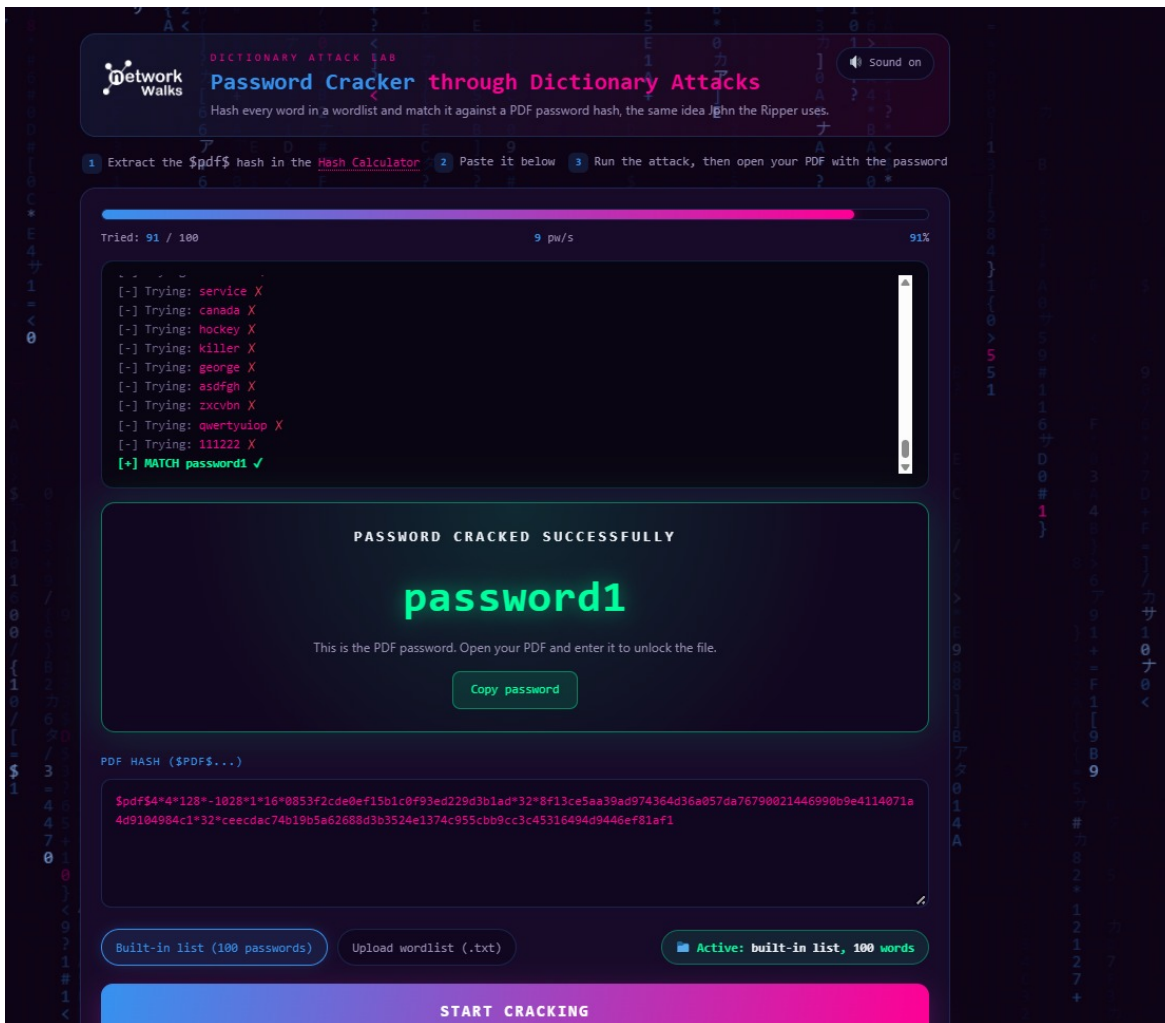
CopyDownload

Revision: R4 Version: V4 Key length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

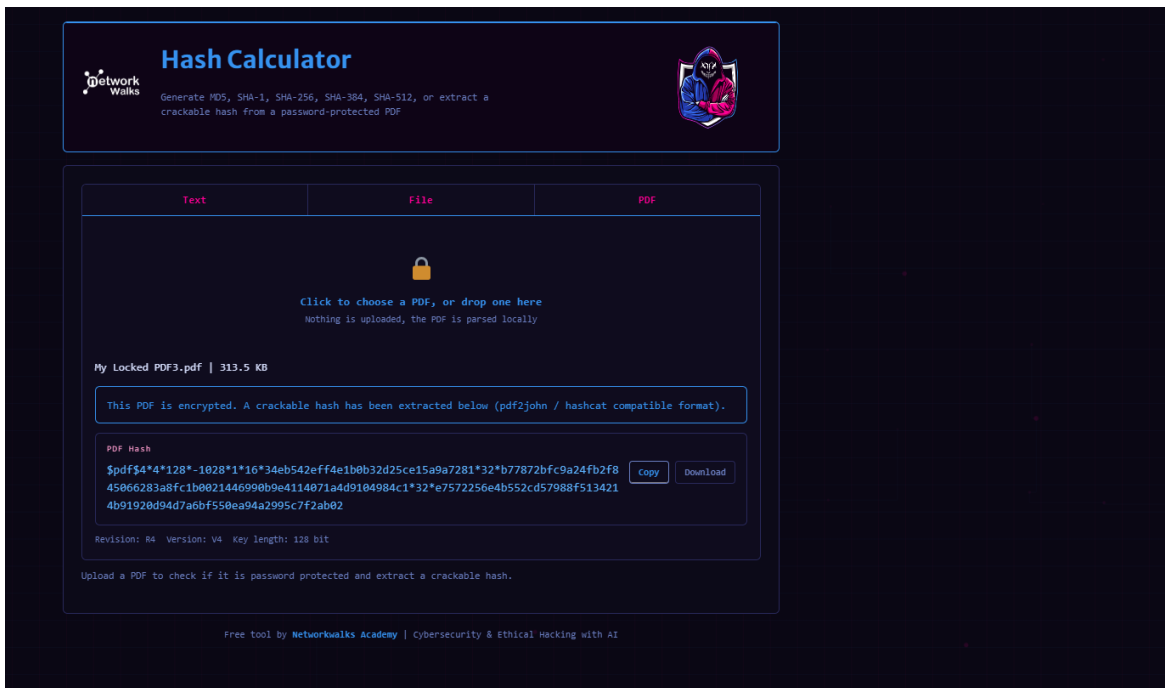
Free tool by Networkwalks Academy | Cybersecurity & Ethical Hacking with AI

E17 — PDF2 crackable hash extracted by Networkwalks.

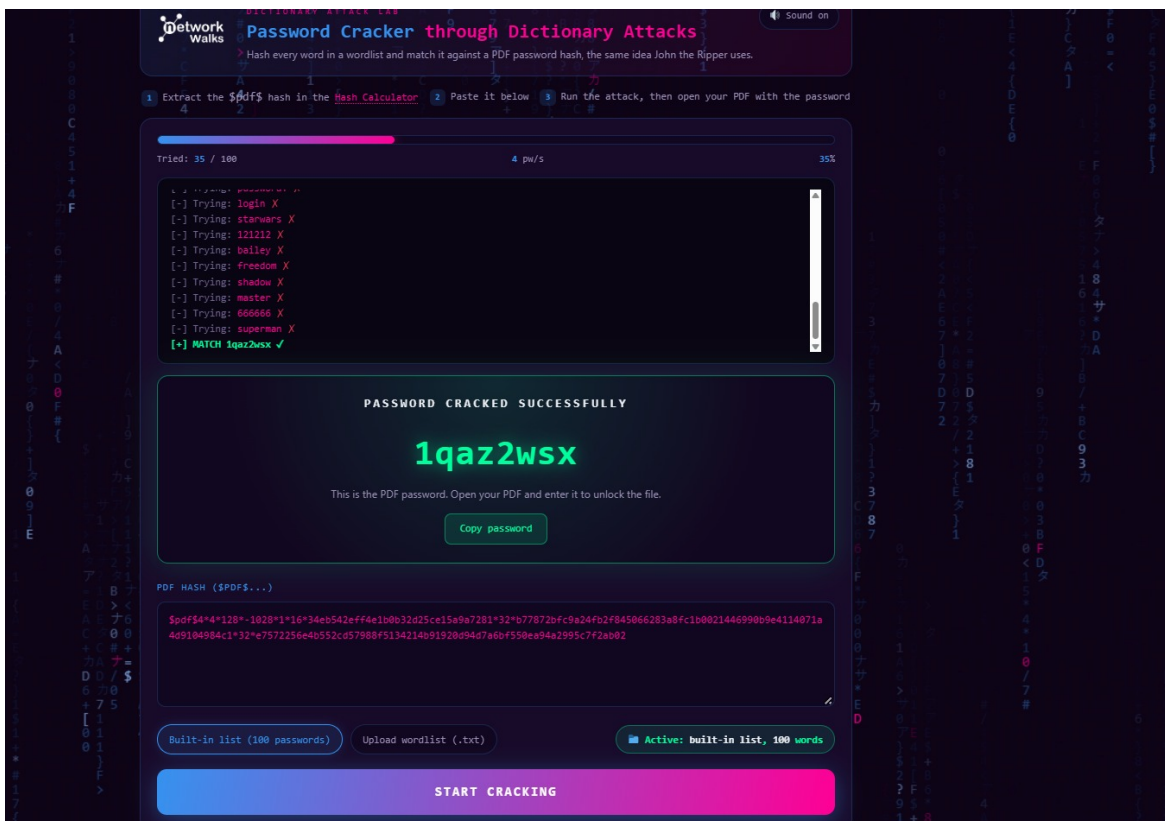


E18 — Networkwalks successfully recovered the PDF2 password after 91/100 candidates at 9 pw/s.

7.4 PDF3 — Hash extraction and recovery



E19 — PDF3 crackable hash extracted by Networkwalks.



E20 — Networkwalks successfully recovered the PDF3 password after 35/100 candidates at 4 pw/s.

7.5 Laboratory completion evidence

The supplied lab screenshots also contain completion/CTF evidence for the individual exercises. These are retained as private evidence and should not be used as public portfolio images because they expose lab-specific completion material.

Evidence	Use
E21 — PDF1 CTF completion	Private assessment evidence
E22 — PDF2 CTF completion	Private assessment evidence
E23 — PDF3 CTF completion	Private assessment evidence

8. Cross-Tool Comparative Analysis

Dimension	John the Ripper	Networkwalks Tools
Execution model	Local CLI	Browser-based
Hash extraction	pdf2john.pl	Networkwalks Hash Calculator
Candidate corpus in this test	RockYou	Built-in 100-password list
Candidate control	High — local wordlists and cracking modes	Lower in built-in mode; interface provides wordlist upload capability
PDF1 outcome	Recovered	Not recovered with captured built-in list
PDF2 outcome	Recovered	Recovered at 91/100, 9 pw/s
PDF3 outcome	Recovered	Recovered at 35/100, 4 pw/s
Operational transparency	Detailed CLI/session/status output	Visual candidate progress and result state
Reproducibility	High when tool version, hash, wordlist and command are preserved	Depends on web-tool behavior, list and interface state
Setup friction	Requires local tooling	Minimal browser setup

8.1 Candidate coverage as the differentiator

PDF1 provides the clearest controlled comparison. The browser tool's captured built-in list was exhausted without a match, while the local RockYou dictionary recovered the password. This means the observed outcome cannot be attributed simply to one engine being capable and the other incapable. The candidate populations were materially different.

8.2 What can and cannot be inferred from speed

The displayed rates—such as 9 pw/s and 4 pw/s in the browser tool and the JtR rates in the local runs—are observations from specific executions. They are not a controlled benchmark because hardware, implementation, candidate ordering, runtime conditions, and execution architecture differ.

8.3 Professional assessment implication

For an analyst, the practical distinction is control and reproducibility. A local cracking engine permits explicit selection of dictionaries, rules, masks, sessions, and logging. A browser tool is useful for rapid educational demonstration, but its result should be interpreted in the context of the wordlist and controls actually exposed by the service.

9. Findings & Security Interpretation

ID	Observation	Evidence	Security interpretation
F-01	Predictable laboratory passwords were recoverable offline	E04/E05/E11/E12 and PM2 results	Once a verifier is exposed, password predictability becomes a primary recovery factor.
F-02	Dictionary coverage materially changed recovery outcome	E16 vs E04/E05	A failed bounded dictionary attack does not prove password strength.
F-03	PDF password-verification material can be converted into a crackable representation	E04/E14/E17/E19	Protection of encrypted artifacts must include protection of the artifacts themselves and control of verifier exposure.
F-04	Browser tooling reduces setup friction but can constrain attack control	E13–E20	Tool convenience and professional assessment control are different requirements.
F-05	Independent validation strengthens a recovery claim	E08–E10	A recovered candidate should be validated against the original protected artifact where feasible.

9.1 Severity framing

No enterprise vulnerability severity is assigned to these laboratory observations because the exercise does not establish a production asset, user population, credential database, exposure path, or business impact. The correct conclusion is a laboratory security observation: dictionary-derived passwords can be recovered when the relevant password verifier is available offline.

10. Defensive Recommendations

Control	Technical recommendation	Why it matters
Password construction	Reject common, breached, predictable and dictionary-derived passwords; favor long unique passphrases.	Directly reduces dictionary-attack success.
Password storage	Use a modern adaptive password hashing scheme with unique salts and appropriately tuned work factors.	Raises offline attacker cost.
MFA	Use phishing-resistant MFA for privileged and high-value access.	Reduces the impact of password compromise alone.
Password reuse	Prevent reuse across organizational systems and known breached credentials.	Limits credential-stuffing and blast radius.
Credential exposure	Treat password-verification databases and encrypted documents as sensitive assets.	Offline access removes online lockout controls.
Monitoring	Detect password spraying, credential stuffing, anomalous authentication and recovery activity.	Improves detection around credential attacks.
Evidence handling	Keep recovered lab credentials out of public repositories and portfolio images.	Prevents secondary disclosure.
Security testing	Preserve tool versions, wordlists, commands, timestamps and raw outputs.	Makes assessments reproducible and auditable.

11. Evidence Quality & Chain of Custody

The report uses stable evidence IDs rather than arbitrary screenshot filenames. Each screenshot has been renamed according to phase, artifact, action and evidence number.

Evidence property	Implementation
Stable identifier	E01–E25
Phase prefix	PM1 / PM2
Artifact context	PDF1 / PDF2 / PDF3 where applicable
Action/result	Hash extraction, recovery, validation, etc.
Private/public separation	Complete evidence set vs Public_Safe set
Index	EVIDENCE_INDEX.md included in evidence pack
Credential handling	Successful recovery screenshots retained privately; public-safe set excludes visible recovered passwords/CTF flags

This structure makes the same evidence usable in three contexts: the formal report, a GitHub project directory, and a LinkedIn project post. The report can cite E-numbers, GitHub can reference stable filenames, and LinkedIn can use the public-safe subset without exposing laboratory credentials.

12. Limitations

- The Networkwalks PDF1 negative result is limited to the captured built-in candidate list; it is not evidence that the password is cryptographically resistant.
- Observed cracking rates are execution-specific and should not be treated as performance benchmarks between the two tools.
- PDF1 has a captured independent qpdf acceptance, decryption, integrity and content-inspection chain. PDF2 and PDF3 recovery is evidenced by JtR and Networkwalks outputs; this report does not claim independent qpdf validation for those two files.
- The assessment corpus contains only three laboratory PDFs and cannot establish statistical conclusions about password populations.
- No production authentication service, user database, or live account was assessed in this exercise.
- CTF/completion screenshots are evidence of laboratory completion, not evidence of production security.

13. Technical Conclusion

The Week 3 assessment demonstrates the full lifecycle of a controlled offline password-recovery exercise. The important professional distinction is between executing a cracking command and proving what the command actually established.

The evidence demonstrates that all three supplied PDFs contained password-verification material that could be processed by the tested cracking workflows. John the Ripper with RockYou recovered all three laboratory passwords. The Networkwalks browser workflow recovered PDF2 and PDF3 using its built-in candidate list, while PDF1 exhausted that bounded list without a match.

PDF1 is the strongest comparative case because the same artifact produced different outcomes under different candidate populations. The result demonstrates that dictionary attacks are a function of both the verifier and the candidate corpus. A negative result from a small wordlist should therefore never be reported as proof of strong password security.

Final assessment position

The laboratory evidence supports a defensive conclusion: organizations should assume that weak or predictable passwords become materially exposed when password-verification material is obtained offline. Strong password selection, adaptive password hashing, MFA, credential uniqueness, sensitive-artifact protection, and disciplined evidence handling form complementary controls.

Appendix A — Evidence Index

ID	Renamed evidence file
E01	E01_PM1_John_the_Ripper_Version_and_Help.png
E02	E02_PM1_John_Manual_Dictionary_Mode.png
E03	E03_PM1_PDF1_Hash_Extraction_and_pdf2john_Discovery.png
E04	E04_PM1_PDF1_Hash_Extraction_and_Format_Validation.png
E05	E05_PM1_RockYou_Wordlist_Preparation_and_JtR_Configuration.png
E06	E06_PM1_QPDF_Installation.png
E07	E07_PM1_QPDF_Installation_and_Reference_Output.png
E08	E08_PM1_PDF1_QPDF_Password_and_AESv2_Validation.png
E09	E09_PM1_PDF1_Decryption_QPDF_Check_and_PDFInfo.png
E10	E10_PM1_PDF1_Content_Inspection_and_Page_Rendering.png
E11	E11_PM1_PDF2_JtR_Recovery_and_Show_Confirmation.png
E12	E12_PM1_PDF3_JtR_Recovery_and_Show_Confirmation.png
E13	E13_PM2_PDF1_Networkwalks_Hash_Calculator_Upload.png
E14	E14_PM2_PDF1_Networkwalks_Hash_Extraction.png
E15	E15_PM2_PDF1_Networkwalks_Cracker_Input.png
E16	E16_PM2_PDF1_Builtin_Dictionary_Exhausted_No_Match.png
E17	E17_PM2_PDF2_Networkwalks_Hash_Extraction.png
E18	E18_PM2_PDF2_Networkwalks_Password_Recovered.png
E19	E19_PM2_PDF3_Networkwalks_Hash_Extraction.png
E20	E20_PM2_PDF3_Networkwalks_Password_Recovered.png
E21	E21_PM2_PDF1_CTF_Completion.png
E22	E22_PM2_PDF2_CTF_Completion.png
E23	E23_PM2_PDF3_CTF_Completion.png
E24	E24_PM1_Kali_Linux_Environment_Preparation.png
E25	E25_PM1_Poppler_Uilities_Installation.png

Appendix B — Reproducible Command Record

```

/usr/share/john/pdf2john.pl "My Locked PDF1.pdf" > pdf1.hash
john --format=pdf --show pdf1.hash
john --format=pdf --wordlist=/usr/share/wordlists/rockyou.txt pdf1.hash
john --format=pdf --show pdf1.hash
qpdf --password='good-luck' --show-encryption "My Locked PDF1.pdf"
qpdf --password='good-luck' --decrypt "My Locked PDF1.pdf" "My Locked PDF1_decrypted.pdf"
qpdf --check "My Locked PDF1_decrypted.pdf"
pdftinfo "My Locked PDF1_decrypted.pdf"
pdftimages -list "My Locked PDF1_decrypted.pdf"
pdftoppm -png -f 1 -singlefile "My Locked PDF1_decrypted.pdf" "My_Locked_PDF1_page1"
/usr/share/john/pdf2john.pl "My Locked PDF2.pdf" > pdf2.hash
john --format=pdf --wordlist=/usr/share/wordlists/rockyou.txt pdf2.hash
john --format=pdf --show pdf2.hash
/usr/share/john/pdf2john.pl "My Locked PDF3.pdf" > pdf3.hash
john --format=pdf --wordlist=/usr/share/wordlists/rockyou.txt pdf3.hash
john --format=pdf --show pdf3.hash

```

PM2 browser workflow: Networkwalks Hash Calculator → upload protected PDF → extract crackable \$pdf\$ hash → Networkwalks Password Cracker → submit hash → run dictionary attack → record candidate progress/result.

Appendix C — Public Portfolio Handling

For GitHub and LinkedIn, use the `Public\_Safe/` evidence subset from the renamed evidence pack. It intentionally excludes screenshots that visibly expose recovered passwords or CTF completion flags.

Use case	Recommended material
Formal assessment report	Complete E01–E25 private evidence set
GitHub repository	Public_Safe images + EVIDENCE_INDEX.md; keep recovered credentials private
LinkedIn post	2–4 Public_Safe technical screenshots; emphasize methodology and candidate coverage
Private interview portfolio	Complete evidence set, subject to lab-owner permission

End of Report